Although the upper limit of integration is infinity, the exponentially decaying envelope ensures $g(t)$ is effectively zero well before $t = 100$. Thus, an upper limit of $t = 100$ is used along with $\Delta t = 0.001$.

```
>>  t = (0:0.001:100);
>>  E_g = sum(g_squared(t)*0.001)
    E_g = 0.2567
```

A slightly better approximation is obtained with the quad function.

```
>>  E_g = quad(g_squared,0,100)
    E_g = 0.2562
```

DRILL 1.21 Computing Signal Energy with MATLAB

Use MATLAB to confirm that the energy of signal $h(t)$, defined previously as $h(t) = g(2t+1) + g(-t+1)$, is $E_h = 0.3768$.

1.12 SUMMARY

A *signal* is a set of data or information. A *system* processes input signals to modify them or extract additional information from them to produce output signals (response). A system may be made up of physical components (hardware realization), or it may be an algorithm that computes an output signal from an input signal (software realization).

A convenient measure of the size of a signal is its energy, if it is finite. If the signal energy is infinite, the appropriate measure is its power, if it exists. The signal power is the time average of its energy (averaged over the entire time interval from $-\infty$ to ∞). For periodic signals, the time averaging need be performed over only one period in view of the periodic repetition of the signal. Signal power is also equal to the mean squared value of the signal (averaged over the entire time interval from $t = -\infty$ to ∞).

Signals can be classified in several ways.

1. A *continuous-time signal* is specified for a continuum of values of the independent variable (such as time t). A *discrete-time signal* is specified only at a finite or a countable set of time instants.
2. An *analog signal* is a signal whose amplitude can take on any value over a continuum. On the other hand, a signal whose amplitudes can take on only a finite number of values is a *digital signal*. The terms *discrete-time* and *continuous-time* qualify the nature of a signal along the time axis (horizontal axis). The terms *analog* and *digital,* on the other hand, qualify the nature of the signal amplitude (vertical axis).
3. A *periodic signal* $x(t)$ is defined by the fact that $x(t) = x(t+T_0)$ for some T_0. The smallest positive value of T_0 for which this relationship is satisfied is called the *fundamental period*. A periodic signal remains unchanged when shifted by an integer multiple of its period. A periodic signal $x(t)$ can be generated by a periodic extension of any contiguous segment of $x(t)$ of duration T_0. Finally, a periodic signal, by definition, must exist over the entire time interval $-\infty < t < \infty$. A signal is *aperiodic* if it is not periodic.

4. An *everlasting signal* starts at $t = -\infty$ and continues forever to $t = \infty$. Hence, periodic signals are everlasting signals. A *causal signal* is a signal that is zero for $t < 0$.
5. A signal with finite energy is an *energy signal*. Similarly a signal with a finite and nonzero power (mean-square value) is a *power signal*. A signal can be either an energy signal or a power signal, but not both. However, there are signals that are neither energy nor power signals.
6. A signal whose physical description is known completely in a mathematical or graphical form is a *deterministic signal*. A *random signal* is known only in terms of its probabilistic description such as mean value or mean-square value, rather than by its mathematical or graphical form.

A signal $x(t)$ delayed by T seconds (right-shifted) can be expressed as $x(t-T)$; on the other hand, $x(t)$ advanced by T (left-shifted) is $x(t+T)$. A signal $x(t)$ time-compressed by a factor $a\,(a > 1)$ is expressed as $x(at)$; on the other hand, the same signal time-expanded by factor $a\,(a > 1)$ is $x(t/a)$. The signal $x(t)$ when time-reversed can be expressed as $x(-t)$.

The unit step function $u(t)$ is very useful in representing causal signals and signals with different mathematical descriptions over different intervals.

In the classical (Dirac) definition, the unit impulse function $\delta(t)$ is characterized by unit area and is concentrated at a single instant $t = 0$. The impulse function has a sampling (or sifting) property, which states that the area under the product of a function with a unit impulse is equal to the value of that function at the instant at which the impulse is located (assuming the function to be continuous at the impulse location). In the modern approach, the impulse function is viewed as a generalized function and is defined by the sampling property.

The exponential function e^{st}, where s is complex, encompasses a large class of signals that includes a constant, a monotonic exponential, a sinusoid, and an exponentially varying sinusoid.

A real signal that is symmetrical about the vertical axis ($t = 0$) is an *even* function of time, and a real signal that is antisymmetrical about the vertical axis is an *odd* function of time. The product of an even function and an odd function is an odd function. However, the product of an even function and an even function or an odd function and an odd function is an even function. The area under an odd function from $t = -a$ to a is always zero regardless of the value of a. On the other hand, the area under an even function from $t = -a$ to a is two times the area under the same function from $t = 0$ to a (or from $t = -a$ to 0). Every signal can be expressed as a sum of odd and even functions of time.

A system processes input signals to produce output signals (response). The input is the cause, and the output is its effect. In general, the output is affected by two causes: the internal conditions of the system (such as the initial conditions) and the external input.

Systems can be classified in several ways.

1. Linear systems are characterized by the linearity property, which implies superposition; if several causes (such as various inputs and initial conditions) are acting on a linear system, the total output (response) is the sum of the responses from each cause, assuming that all the remaining causes are absent. A system is nonlinear if superposition does not hold.
2. In time-invariant systems, system parameters do not change with time. The parameters of time-varying-parameter systems change with time.
3. For memoryless (or instantaneous) systems, the system response at any instant t depends only on the value of the input at t. For systems with memory (also known as dynamic

systems), the system response at any instant t depends not only on the present value of the input, but also on the past values of the input (values before t).

4. In contrast, if a system response at t also depends on the future values of the input (values of input beyond t), the system is noncausal. In causal systems, the response does not depend on the future values of the input. Because of the dependence of the response on the future values of input, the effect (response) of noncausal systems occurs before the cause. When the independent variable is time (temporal systems), the noncausal systems are prophetic systems, and therefore, unrealizable, although close approximation is possible with some time delay in the response. Noncausal systems with independent variables other than time (e.g., space) are realizable.
5. Systems whose inputs and outputs are continuous-time signals are continuous-time systems; systems whose inputs and outputs are discrete-time signals are discrete-time systems. If a continuous-time signal is sampled, the resulting signal is a discrete-time signal. We can process a continuous-time signal by processing the samples of the signal with a discrete-time system.
6. Systems whose inputs and outputs are analog signals are analog systems; those whose inputs and outputs are digital signals are digital systems.
7. If we can obtain the input $x(t)$ back from the output $y(t)$ of a system $\mathcal{S}$ by some operation, the system $\mathcal{S}$ is said to be invertible. Otherwise the system is noninvertible.
8. A system is stable if bounded input produces bounded output. This defines external stability because it can be ascertained from measurements at the external terminals of the system. External stability is also known as the stability in the BIBO (bounded-input/bounded-output) sense. Internal stability, discussed later in Ch. 2, is measured in terms of the internal behavior of the system.

The system model derived from a knowledge of the internal structure of the system is its internal description. In contrast, an external description is a representation of a system as seen from its input and output terminals; it can be obtained by applying a known input and measuring the resulting output. In the majority of practical systems, an external description of a system so obtained is equivalent to its internal description. At times, however, the external description fails to describe the system adequately. Such is the case with the so-called uncontrollable or unobservable systems.

A system may also be described in terms of certain set of key variables called state variables. In this description, an Nth-order system can be characterized by a set of N simultaneous first-order differential equations in N state variables. State equations of a system represent an internal description of that system.

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PROBLEMS

1.1-1 Find the energies of the signals illustrated in Fig. P1.1-1. Comment on the effect on energy of sign change, time shifting, or doubling of the signal. What is the effect on the energy if the signal is multiplied by k?

1.1-2 Repeat Prob. 1.1-1 for the signals in Fig. P1.1-2.

1.1-3 (a) Find the energies of the pair of signals $x(t)$ and $y(t)$ depicted in Figs. P1.1-3a and P1.1-3b. Sketch and find the energies of signals $x(t) + y(t)$ and $x(t) - y(t)$. Can you make any observation from these results?

(b) Repeat part (a) for the signal pair illustrated in Fig. P1.1-3c. Is your observation in part (a) still valid?

1.1-4 Find the power of the periodic signal $x(t)$ shown in Fig. P1.1-4. Find also the powers and the rms values of:

(a) $-x(t)$
(b) $2x(t)$
(c) $cx(t)$

Comment.

1.1-5 By original design, a system outputs a 10-volt pulse that is 3 seconds in duration. It is desired to upgrade the square-pulse output with a "soft-start" pulse that steps up to 10 volts in 1-volt increments spaced every 20 milliseconds. Determine the signal duration T so that the "soft-start" pulse has the same signal energy as the original square pulse.

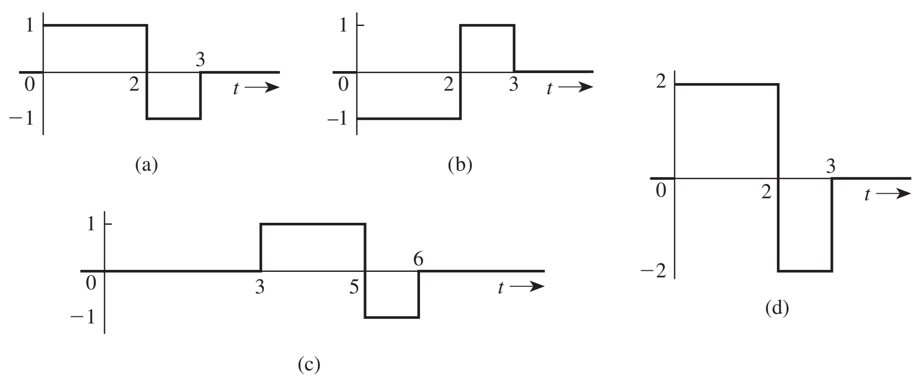

Figure P1.1-1

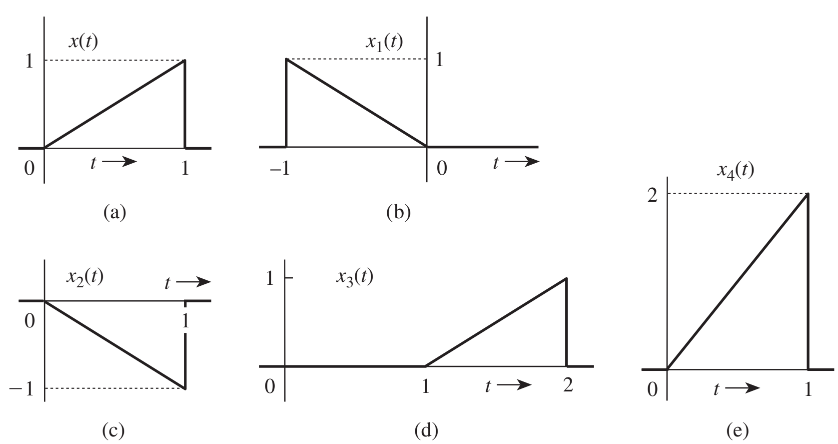

Figure P1.1-2

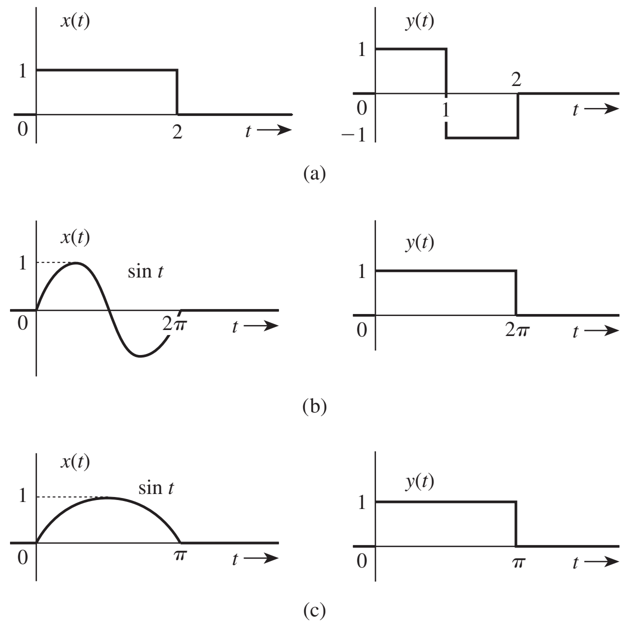

Figure P1.1-3

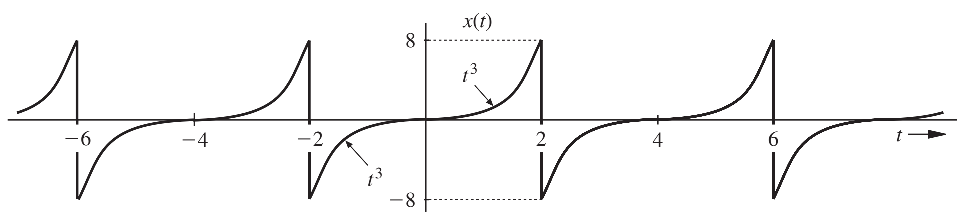

Figure P1.1-4

1.1-6 Determine the power and the rms value for each of the following signals:
(a) $5+10\cos(100t+\pi/3)$
(b) $10\cos(100t+\pi/3)+16\sin(150t+\pi/5)$
(c) $(10+2\sin 3t)\cos 10t$
(d) $10\cos 5t\cos 10t$
(e) $10\sin 5t\cos 10t$
(f) $e^{j\alpha t}\cos\omega_0 t$

1.1-7 Figure P1.1-7 shows a periodic 50% duty cycle dc-offset sawtooth wave $x(t)$ with peak amplitude A. Determine the energy and power of $x(t)$.

1.1-8 Two periodic signals that differ only by a 90-degree phase shift are considered to be quadrature signals. For example, $\cos(2\pi t)$ and $\sin(2\pi t)$ are quadrature signals. Another pair of quadrature signals is $x(t) = \text{sgn}[\cos(2\pi t)]$ and $y(t) = \text{sgn}[\sin(2\pi t)]$, where sgn is the sign (or signum) function.
(a) Plot $x(t)$ and determine its power P_x and energy E_x.
(b) Plot $y(t)$ and determine its power P_y and energy E_y.
(c) Consider the complex function $f(t) = x(t) + jy(t)$. Determine the power and energy of $f(t)$.
(d) When real functions $x(t)$ and $y(t)$ are combined as $f(t) = x(t) + jy(t)$, is it generally true that $E_f = E_x + E_y$ and $P_f = P_x + P_y$? Prove your answer.

1.1-9 There are many useful properties related to signal energy. Prove each of the following statements. In each case, let energy signal $x_1(t)$ have energy $E[x_1(t)]$, let energy signal $x_2(t)$ have

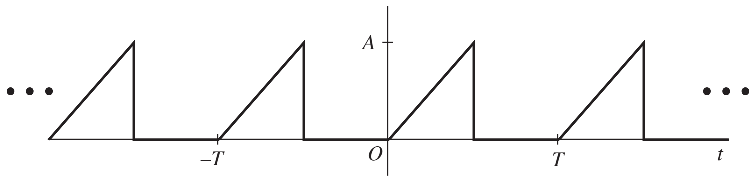

Figure P1.1-7

energy $E[x_2(t)]$, and let T be a nonzero, finite, real-valued constant.

(a) Prove $E[Tx_1(t)] = T^2E[x_1(t)]$. That is, amplitude scaling a signal by constant T scales the signal energy by T^2.

(b) Prove $E[x_1(t)] = E[x_1(t - T)]$. That is, shifting a signal does not affect its energy.

(c) If $(x_1(t) \neq 0) \Rightarrow (x_2(t) = 0)$ and $(x_2(t) \neq 0) \Rightarrow (x_1(t) = 0)$, then prove $E[x_1(t) + x_2(t)] = E[x_1(t)] + E[x_2(t)]$. That is, the energy of the sum of two nonoverlapping signals is the sum of the two individual energies.

(d) Prove $E[x_1(Tt)] = (1/|T|)E[x_1(t)]$. That is, time-scaling a signal by T reciprocally scales the signal energy by $1/|T|$.

1.1-10 Consider the signal $x(t)$ shown in Fig. P1.1-10. Outside the interval shown, $x(t)$ is zero. Determine the signal energy $E[x(t)]$. [*Hint:* Use the results of Prob. 1.1-9.]

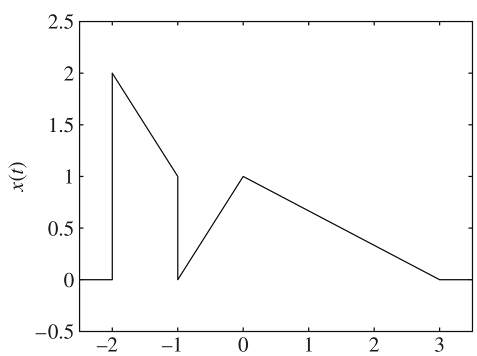

Figure P1.1-10

1.1-11 (a) Show that the power of a signal

$$x(t) = \sum_{k=m}^{n} D_k e^{j\omega_k t}$$

is

$$P_x = \sum_{k=m}^{n} |D_k|^2$$

assuming all frequencies to be distinct, that is, $\omega_i \neq \omega_k$ for all $i \neq k$.

(b) Use the result in part (a) to determine the power of each of the signals in Prob. 1.1-6.

1.1-12 A binary signal $x(t) = 0$ for $t < 0$. For positive time, $x(t)$ toggles between one and zero as follows: one for 1 second, zero for 1 second, one for 1 second, zero for 2 seconds, one for 1 second, zero for 3 seconds, and so forth. That is, the "on" time is always 1 second, but the "off" time successively increases by 1 second between each toggle. A portion of $x(t)$ is shown in Fig. P1.1-12. Determine the energy and power of $x(t)$.

1.2-1 For the signal $x(t)$ depicted in Fig. P1.2-1, sketch the signals
(a) $x(-t)$
(b) $x(t+6)$
(c) $x(3t)$
(d) $x(t/2)$

1.2-2 For the signal $x(t)$ illustrated in Fig. P1.2-2, sketch
(a) $x(t-4)$
(b) $x(t/1.5)$
(c) $x(-t)$
(d) $x(2t-4)$
(e) $x(2-t)$

1.2-3 In Fig. P1.2-3, express signals $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$, and $x_5(t)$ in terms of signal $x(t)$ and its time-shifted, time-scaled, or time-reversed versions.

1.2-4 For an energy signal $x(t)$ with energy E_x, show that the energy of any one of the signals $-x(t)$, $x(-t)$, and $x(t-T)$ is E_x. Show also that the energy of $x(at)$ as well as $x(at-b)$ is E_x/a, but the energy of $ax(t)$ is a^2E_x. This shows that time

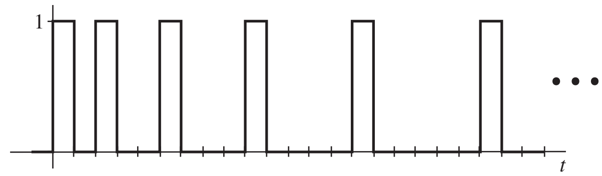

Figure P1.1-12

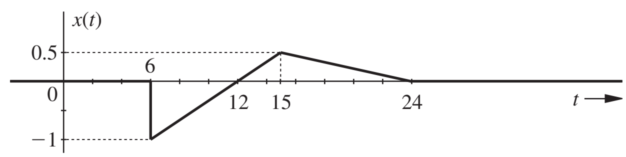

Figure P1.2-1

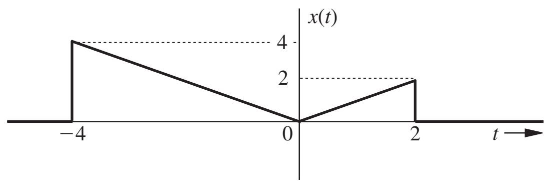

Figure P1.2-2

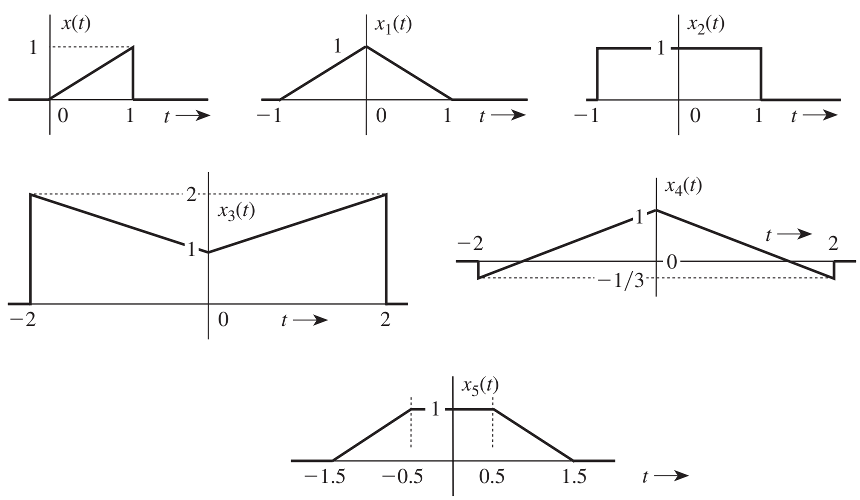

Figure P1.2-3

inversion and time shifting do not affect signal energy. On the other hand, time compression of a signal ($a > 1$) reduces the energy, and time expansion of a signal ($a < 1$) increases the energy. What is the effect on signal energy if the signal is multiplied by a constant a?

1.2-5 Define $2x(-3t+1) = t[u(-t-1) - u(-t+1)]$, where $u(t)$ is the unit step function.

(a) Plot $2x(-3t+1)$ over a suitable range of t.

(b) Plot $x(t)$ over a suitable range of t.

1.2-6 Consider the signal $x(t) = 2^{-tu(t)}$, where $u(t)$ is the unit step function.

(a) Accurately sketch $x(t)$ over $(-1 \leq t \leq 1)$.

(b) Accurately sketch $y(t) = 0.5x(1-2t)$ over $(-1 \leq t \leq 1)$.

1.2-7 Define signals $y(t)$ and $z(t)$ as in Fig. P1.2-7.

(a) Determine constants a, b, and c to produce $z(t) = ax(bt+c)$ in Fig. P1.2-7.

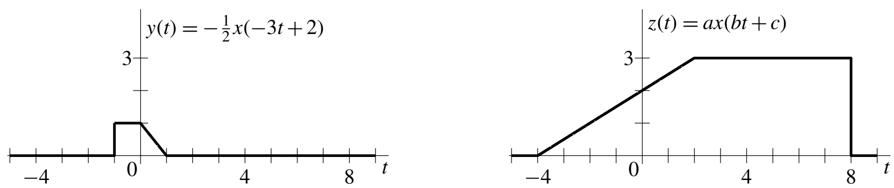

Figure P1.2-7

(b) Determine and sketch a signal $v(t)$ such that $z(t) = \int_{-\infty}^{t} v(\tau)\,d\tau$.

1.3-1 Think of a real-world signal that is a personally relevant and interesting. Describe the signal and then classify it according to the six following characteristics:

(a) continuous-time or discrete-time
(b) analog or digital
(c) periodic or aperiodic
(d) energy or power
(e) causal or noncausal
(f) deterministic or random

If possible, think of a second real-world signal that has the opposite six characteristics of your first signal. If such a second signal is not possible, carefully explain why that is the case.

1.3-2 Define signal $y(t) = \sum_{k=-\infty}^{\infty} x(0.5t - 10k)$, where

$$x(t) = \begin{cases} e^{-2t} & t \geq 1 \\ 0 & t < 1 \end{cases}$$

(a) Determine the constant a such that the signal $x(-2t+a)$ is borderline anticausal.
(b) Is the signal $y(t)$ periodic? If so, determine the period T_y. If not, explain why $y(t)$ is not periodic.

1.3-3 Determine whether each of the following statements is true or false. If the statement is false, demonstrate this by proof or example.

(a) Every continuous-time signal is an analog signal.
(b) Every discrete-time signal is a digital signal.
(c) If a signal is not an energy signal, then it must be a power signal and vice versa.
(d) An energy signal must be of finite duration.
(e) A power signal cannot be causal.
(f) A periodic signal cannot be anticausal.

1.3-4 Determine whether each of the following statements is true or false. If the statement is false, demonstrate by proof or example why the statement is false.

(a) Every bounded periodic signal is a power signal.
(b) Every bounded power signal is a periodic signal.
(c) If an energy signal $x(t)$ has energy E, then the energy of $x(at)$ is E/a. Assume a is real and positive.
(d) If a power signal $x(t)$ has power P, then the power of $x(at)$ is P/a. Assume a is real and positive.

1.3-5 Given $x_1(t) = \cos(t)$, $x_2(t) = \sin(\pi t)$, and $x_3(t) = x_1(t) + x_2(t)$.

(a) Determine the fundamental periods T_1 and T_2 of signals $x_1(t)$ and $x_2(t)$.
(b) Show that $x_3(t)$ is not periodic, which requires $T_3 = k_1 T_1 = k_2 T_2$ for some integers k_1 and k_2.
(c) Determine the powers P_{x_1}, P_{x_2}, and P_{x_3} of signals $x_1(t)$, $x_2(t)$, and $x_3(t)$.

1.3-6 For any constant ω, is the function $f(t) = \sin(\omega t)$ a periodic function of the independent variable t? Justify your answer.

1.3-7 The signal shown in Fig. P1.3-7 is defined as

$$x(t) = \begin{cases} t & 0 \leq t < 1 \\ 0.5 + 0.5\cos(2\pi t) & 1 \leq t < 2 \\ 3 - t & 2 \leq t < 3 \\ 0 & \text{otherwise} \end{cases}$$

The energy of $x(t)$ is $E \approx 1.0417$.

(a) What is the energy of $y_1(t) = (1/3)x(2t)$?
(b) A periodic signal $y_2(t)$ is defined as

$$y_2(t) = \begin{cases} x(t) & 0 \leq t < 4 \\ y_2(t+4) & \forall t \end{cases}$$

What is the power of $y_2(t)$?

(c) What is the power of $y_3(t) = (1/3)y_2(2t)$?

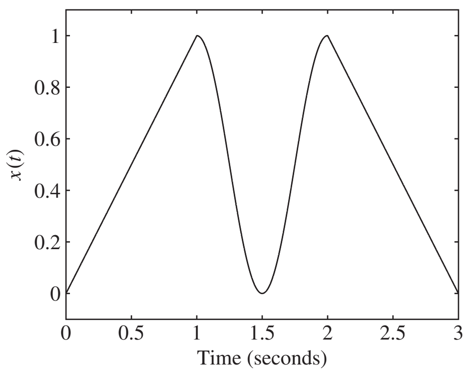

Figure P1.3-7

1.3-8 Let $y_1(t) = y_2(t) = t^2$ over $0 \le t \le 1$. Notice, this statement does not require $y_1(t) = y_2(t)$ for all t.

(a) Define $y_1(t)$ as an even, periodic signal with period $T_1 = 2$. Sketch $y_1(t)$ and determine its power.

(b) Design an odd, periodic signal $y_2(t)$ with period $T_2 = 3$ and power equal to unity. Fully describe $y_2(t)$ and sketch the signal over at least one full period. [*Hint:* There are an infinite number of possible solutions to this problem—you need to find only one of them!]

(c) We can create a complex-valued function $y_3(t) = y_1(t) + jy_2(t)$. Determine whether this signal is periodic. If yes, determine the period T_3. If no, justify why the signal is not periodic.

(d) Determine the power of $y_3(t)$ defined in part (c). The power of a complex-valued function $z(t)$ is

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} z(\tau)z^*(\tau)\,d\tau$$

1.4-1 Sketch the following signals:

(a) $u(t-5) - u(t-7)$

(b) $u(t-5) + u(t-7)$

(c) $t^2[u(t-1) - u(t-2)]$

(d) $(t-4)[u(t-2) - u(t-4)]$

1.4-2 Express each of the signals in Fig. P1.4-2 by a single expression valid for all t.

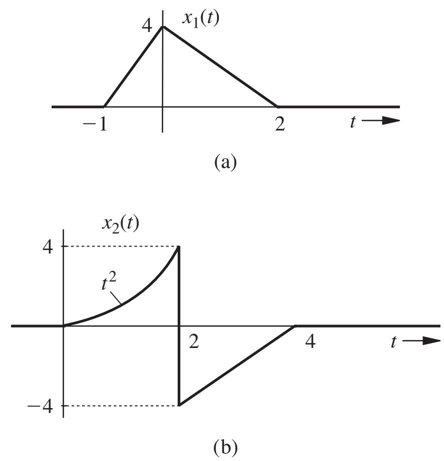

Figure P1.4-2

1.4-3 Letting $w(t) = t\,[u(t) - u(t-1)]$, define the periodic signal $x(t)$ as

$$x(t) = \sum_{k=-\infty}^{\infty} w(2t+2k) - 0.5w(2t+2k-1)$$

(a) Sketch $w(t)$ and $x(t)$. What is the fundamental period T_0 of signal $x(t)$?

(b) Sketch $y(t) = \frac{d}{dt}x(1-0.5t)$.

(c) Determine the energy E_z and power P_z of the signal $z(t) = x(0.5 - 1.5t)\,[u(t) - u(t-1)]$. Sketching $z(t)$ should help.

1.4-4 Define signal $x(t) = u(t-1) - u(t-2.5) - 2\delta(t-4) + \delta(t-6)$.

(a) Sketch $y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$.

(b) Describe a simple change that can be made to the right-most delta function in $x(t)$ so that $y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$ has finite energy.

(c) Sketch $z(t) = \int_{t}^{\infty} x(\tau)\,d\tau$.

(d) Determine real constants A and B so that $w(t) = x\left(\frac{t-A}{B}\right)$ has a region of support $[-2, 2]$.

1.4-5 Simplify the following expressions:

(a) $\left(\dfrac{\sin t}{t^2+2}\right)\delta(t)$

(b) $\left(\dfrac{j\omega+2}{\omega^2+9}\right)\delta(\omega)$

(c) $[e^{-t}\cos(3t - 60°)]\delta(t)$

(d) $\left(\dfrac{\sin\left[\frac{\pi}{2}(t-2)\right]}{t^2+4}\right)\delta(1-t)$

(e) $\left(\dfrac{1}{j\omega+2}\right)\delta(\omega+3)$

(f) $\left(\dfrac{\sin k\omega}{\omega}\right)\delta(\omega)$

[*Hint:* Use Eq. (1.10). For part (f) use L'Hôpital's rule.]

1.4-6 Evaluate the following integrals:

(a) $\displaystyle\int_{-\infty}^{\infty} \delta(\tau)x(t-\tau)\,d\tau$

(b) $\displaystyle\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$

(c) $\displaystyle\int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}\,dt$

(d) $\displaystyle\int_{-\infty}^{\infty} \delta(2t-3)\sin \pi t\,dt$

(e) $\displaystyle\int_{-\infty}^{\infty} \delta(t+3)e^{-t}\,dt$

(f) $\displaystyle\int_{-\infty}^{\infty} (t^3+4)\delta(1-t)\,dt$

(g) $\displaystyle\int_{-\infty}^{\infty} x(2-t)\delta(3-t)\,dt$

(h) $\displaystyle\int_{-\infty}^{\infty} e^{(x-1)}\cos\left[\tfrac{\pi}{2}(x-5)\right]\delta(x-3)\,dx$

1.4-7 For real and positive constant a, evaluate the following integral:

$$\int_{-\infty}^{\infty} \delta(at)\,dt$$

1.4-8 (a) Find and sketch dx/dt for the signal $x(t)$ shown in Fig. P1.2-2.
(b) Find and sketch d^2x/dt^2 for the signal $x_1(t)$ depicted in Fig. P1.4-2a.

1.4-9 Find and sketch $\int_{-\infty}^{t} x(t)\,dt$ for the signals $x(t)$ illustrated in Fig. P1.4-9.

1.4-10 Using the generalized function definition of impulse [Eq. (1.11) with $T=0$], show that $\delta(t)$ is an even function of t.

1.4-11 Using the generalized function definition of impulse [Eq. (1.11) with $T=0$], show that

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

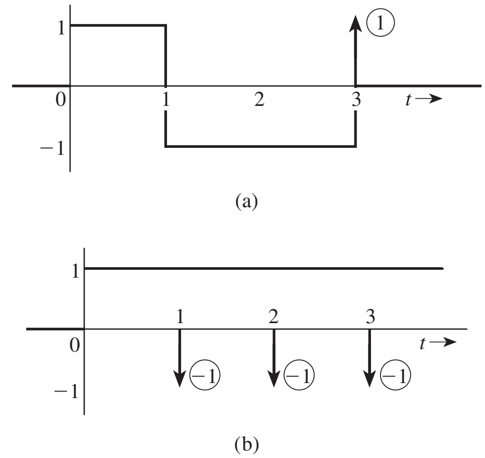

Figure P1.4-9

1.4-12 Show that

$$\int_{-\infty}^{\infty} \dot{\delta}(t)\phi(t)\,dt = -\dot{\phi}(0)$$

where $\phi(t)$ and $\dot{\phi}(t)$ are continuous at $t=0$, and $\phi(t)\to 0$ as $t\to\pm\infty$. This integral defines $\dot{\delta}(t)$ as a generalized function. [*Hint:* Use integration by parts.]

1.4-13 A sinusoid $e^{\sigma t}\cos\omega t$ can be expressed as a sum of exponentials e^{st} and e^{-st} [Eq. (1.14)] with complex frequencies $s=\sigma+j\omega$ and $s=\sigma-j\omega$. Locate in the complex plane the frequencies of the following sinusoids:

(a) $\cos 3t$
(b) $e^{-3t}\cos 3t$
(c) $e^{2t}\cos 3t$
(d) e^{-2t}
(e) e^{2t}
(f) 5

1.5-1 Find and sketch the odd and the even components of the following:

(a) $u(t)$
(b) $tu(t)$
(c) $\sin\omega_0 t$
(d) $\cos\omega_0 t$
(e) $\cos(\omega_0 t+\theta)$
(f) $\sin\omega_0 t u(t)$
(g) $\cos\omega_0 t u(t)$

1.5-2 Define $x(t)=2u(t+1)-u(t-2)-u(t-3)$.
(a) Letting $x_o(t)$ designate the odd portion of $x(t)$, accurately sketch $x_o(1-2t)$.

(b) Letting $x_e(t)$ designate the even portion of $x(t)$, accurately sketch $x_e(2+t/3)$.

1.5-3 (a) Determine even and odd components of the signal $x(t) = e^{-2t}u(t)$.

(b) Show that the energy of $x(t)$ is the sum of energies of its odd and even components found in part (a).

(c) Generalize the result in part (b) for any finite energy signal.

1.5-4 (a) If $x_e(t)$ and $x_o(t)$ are even and the odd components of a real signal $x(t)$, then show that

$$\int_{-\infty}^{\infty} x_e(t)x_o(t)\,dt = 0$$

(b) Show that

$$\int_{-\infty}^{\infty} x(t)\,dt = \int_{-\infty}^{\infty} x_e(t)\,dt$$

1.5-5 An aperiodic signal is defined as $x(t) = \sin(\pi t)u(t)$, where $u(t)$ is the continuous-time step function. Is the odd portion of this signal, $x_o(t)$, periodic? Justify your answer.

1.5-6 An aperiodic signal is defined as $x(t) = \cos(\pi t)u(t)$, where $u(t)$ is the continuous-time step function. Is the even portion of this signal, $x_e(t)$, periodic? Justify your answer.

1.5-7 Consider the signal $x(t)$ shown in Fig. P1.5-7.

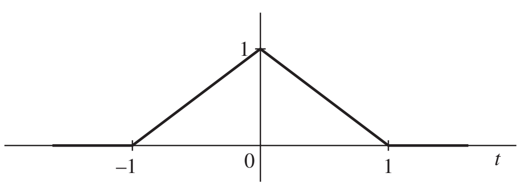

Figure P1.5-7

(a) Determine and carefully sketch $v(t) = 3x(-(1/2)(t+1))$.

(b) Determine the energy and power of $v(t)$.

(c) Determine and carefully sketch the even portion of $v(t)$, $v_e(t)$.

(d) Let $a = 2$ and $b = 3$; sketch $v(at+b)$, $v(at)+b$, $av(t+b)$, and $av(t)+b$.

(e) Let $a = -3$ and $b = -2$; sketch $v(at+b)$, $v(at)+b$, $av(t+b)$, and $av(t)+b$.

1.5-8 Consider the signal $y(t) = (1/5)x(-2t-3)$ shown in Fig. P1.5-8.

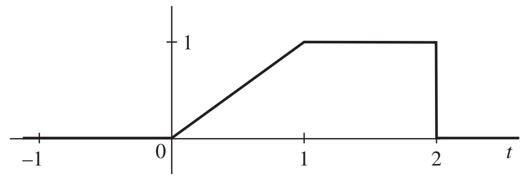

Figure P1.5-8

(a) Does $y(t)$ have an odd portion, $y_o(t)$? If so, determine and carefully sketch $y_o(t)$. Otherwise, explain why no odd portion exists.

(b) Determine and carefully sketch the original signal $x(t)$.

1.5-9 Consider the signal $-(1/2)x(-3t+2)$ shown in Fig. P1.5-9.

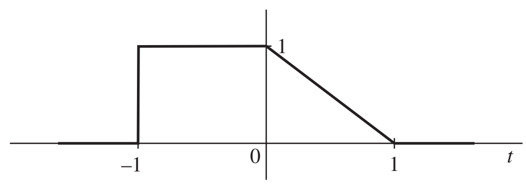

Figure P1.5-9

(a) Determine and carefully sketch the original signal $x(t)$.

(b) Determine and carefully sketch the even portion of the original signal $x(t)$.

(c) Determine and carefully sketch the odd portion of the original signal $x(t)$.

1.5-10 The conjugate symmetric (or Hermitian) portion of a signal is defined as $w_{cs}(t) = (w(t) + w^*(-t))/2$. Show that the real portion of $w_{cs}(t)$ is even and that the imaginary portion of $w_{cs}(t)$ is odd.

1.5-11 The conjugate antisymmetric (or skew-Hermitian) portion of a signal is defined as $w_{ca}(t) = (w(t) - w^*(-t))/2$. Show that the real portion of $w_{ca}(t)$ is odd and that the imaginary portion of $w_{ca}(t)$ is even.

1.5-12 Define $w(t) = e^{j(t+\pi/4)}$.

(a) Referring to the definition in Prob. 1.5-10, determine $w_{cs}(t)$. Express your simplified answer in standard rectangular form.

(b) Referring to the definition in Prob. 1.5-11, determine $w_{ca}(t)$. Express your simplified answer in standard polar form.

1.5-13 Figure P1.5-13 plots a complex signal $w(t)$ in the complex plane over the time range $(0 \leq t \leq 1)$. The time $t=0$ corresponds with the origin, while the time $t=1$ corresponds with the point (2, 1).

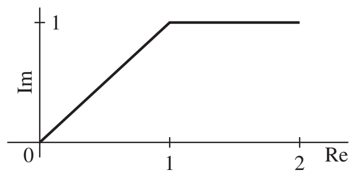

Figure P1.5-13

(a) In the complex plane, plot $w(t)$ over $(-1 \leq t \leq 1)$ if $w(t)$ is an even signal.
(b) In the complex plane, plot $w(t)$ over $(-1 \leq t \leq 1)$ if $w(t)$ is an odd signal.
(c) In the complex plane, plot $w(t)$ over $(-1 \leq t \leq 1)$ if $w(t)$ is a conjugate symmetric signal. [*Hint:* See Prob. 1.5-10.]
(d) In the complex plane, plot $w(t)$ over $(-1 \leq t \leq 1)$ if $w(t)$ is a conjugate antisymmetric signal. [*Hint:* See Prob. 1.5-11.]
(e) In the complex plane, plot as much of $w(3t)$ as possible.

1.5-14 Define complex signal $x(t) = t^2(1+j)$ over interval $(1 \leq t \leq 2)$. The remaining portion is defined such that $x(t)$ is a minimum-energy, skew-Hermitian signal.
(a) Fully describe $x(t)$ for all t.
(b) Sketch $y(t) = \text{Re}\{x(t)\}$ versus the independent variable t.
(c) Sketch $z(t) = \text{Re}\{jx(-2t+1)\}$ versus the independent variable t.
(d) Determine the energy and power of $x(t)$.
[*Hint:* See Prob. 1.5-11 for a definition of skew-Hermitian signals.]

1.6-1 Write the input–output relationship for an ideal integrator. Determine the zero-input and zero-state components of the response.

1.6-2 A force $x(t)$ acts on a ball of mass M (Fig. P1.6-2). Show that the velocity $v(t)$ of the ball at any instant $t > 0$ can be determined if we know the force $x(t)$ over the interval from 0 to t and the ball's initial velocity $v(0)$.

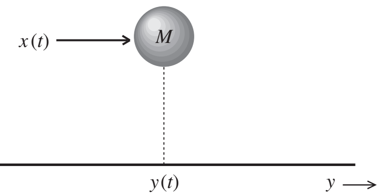

Figure P1.6-2

1.6-3 From your personal experience, provide an example of:
(a) a single-input, single-output (SISO) system
(b) a multiple-input, single-output (MISO) system
(c) a single-input, multiple-output (SIMO) system
(d) a multiple-input, multiple-output (MIMO) system

1.7-1 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, determine which of the systems are linear and which are nonlinear.

(a) $\dfrac{dy(t)}{dt} + 2y(t) = x^2(t)$
(b) $\dfrac{dy(t)}{dt} + 3ty(t) = t^2x(t)$
(c) $3y(t) + 2 = x(t)$
(d) $\dfrac{dy(t)}{dt} + y^2(t) = x(t)$
(e) $\left(\dfrac{dy(t)}{dt}\right)^2 + 2y(t) = x(t)$
(f) $\dfrac{dy(t)}{dt} + (\sin t)y(t) = \dfrac{dx(t)}{dt} + 2x(t)$
(g) $\dfrac{dy(t)}{dt} + 2y(t) = x(t)\dfrac{dx(t)}{dt}$
(h) $y(t) = \displaystyle\int_{-\infty}^{t} x(\tau)\,d\tau$

1.7-2 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, explain with reasons which of the systems are time-invariant parameter systems and which are time-varying-parameter systems.
(a) $y(t) = x(t-2)$
(b) $y(t) = x(-t)$
(c) $y(t) = x(at)$
(d) $y(t) = t\,x(t-2)$
(e) $y(t) = \displaystyle\int_{-5}^{5} x(\tau)\,d\tau$
(f) $y(t) = \left(\dfrac{dx(t)}{dt}\right)^2$

1.7-3 Two inputs, temperature $T(t)$ and wind speed $V(t)$, produce an output, wind chill $W(t)$, according to $W(t) = 35.74 + 0.6215T(t) - 35.75\{V(t)\}^{0.16} + 0.4275T(t)\{V(t)\}^{0.16}$. The independent variable here is time, t. Answer the following questions yes or no, and provide mathematical justification for each answer.

(a) Is this system BIBO-stable?
(b) Is the system memoryless?
(c) Is the system causal?
(d) For simplicity, let the wind speed be constant, $V(t) = k_V$. Thus, $W(t) = k_1 + k_2T(t)$ for some constants k_1 and k_2. Is this simplified system linear?
(e) For simplicity, let the temperature be constant, $T(t) = k_T$. Thus, $W(t) = k_3 + k_4\{V(t)\}^{0.16}$ for some constants k_3 and k_4. Is this simplified system linear?

1.7-4 Input voltage $x(t)$ applied to an inverting op-amp follower circuit produces output $y(t)$ according to

$$y(t+t_{\rm p}) = \begin{cases} -V_{\rm ref} & x(t) > V_{\rm ref} \\ V_{\rm ref} & x(t) < -V_{\rm ref} \\ -x(t) & \text{otherwise} \end{cases}$$

where op-amp reference voltage $V_{\rm ref}$ and propagation delay $t_{\rm p}$ are both positive constants. Answer the following questions yes or no, and provide mathematical justification for each answer.

(a) Is this system BIBO-stable?
(b) Is the system causal?
(c) Is the system invertible?
(d) Is the system linear?
(e) Is the system memoryless?
(f) Is the system time invariant?

1.7-5 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t+1) = \begin{cases} -2x(t) & \text{when } x(t) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

1.7-6 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t-1) = \begin{cases} x(t-1) & \text{when } \frac{d}{dt}x(t) \geq 0 \\ x(t-2) & \text{otherwise} \end{cases}$$

1.7-7 Repeat Prob. 1.7-4 for a system that multiplies a given input by a ramp function, $r(t) = tu(t)$. That is, $y(t) = x(t)r(t)$.

1.7-8 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t) = \frac{d}{dt}x(t-1)$$

1.7-9 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t) = \begin{cases} x(t) & \text{if } \; x(t) > 0 \\ 0 & \text{if } \; x(t) \leq 0 \end{cases}$$

1.7-10 A continuous-time system is given by

$$y(t) = 0.5\int_{-\infty}^{\infty} x(\tau)[\delta(t-\tau) - \delta(t+\tau)]\,d\tau$$

Recall that $\delta(t)$ designates the Dirac delta function.

(a) Explain what this system does.
(b) Is the system BIBO-stable? Justify your answer.
(c) Is the system linear? Justify your answer.
(d) Is the system memoryless? Justify your answer.
(e) Is the system causal? Justify your answer.
(f) Is the system time invariant? Justify your answer.

1.7-11 For a certain LTI system with the input $x(t)$, the output $y(t)$ and the two initial conditions $q_1(0)$ and $q_2(0)$, the following observations were made:

$x(t)$	$q_1(0)$	$q_2(0)$	$y(t)$
0	1	-1	$e^{-t}u(t)$
0	2	1	$e^{-t}(3t+2)u(t)$
$u(t)$	-1	-1	$2u(t)$

Determine $y(t)$ when both the initial conditions are zero and the input $x(t)$ is as shown in Fig. P1.7-11. [*Hint:* There are three causes: the input and each of the two initial conditions. Because of the linearity property, if a cause is increased by a factor k, the response to that cause also increases by the same factor k. Moreover, if causes are added, the corresponding responses add.]

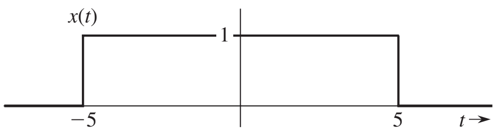

Figure P1.7-11

1.7-12 A system is specified by its input–output relationship as

$$y(t) = \frac{x^2(t)}{dx(t)/dt}$$

Show that the system satisfies the homogeneity property but not the additivity property.

1.7-13 Show that the circuit in Fig. P1.7-13 is zero-state linear but not zero-input linear. Assume all diodes to have identical (matched) characteristics. The output is the current $y(t)$.

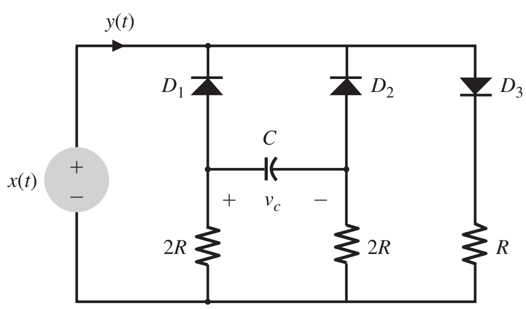

Figure P1.7-13

1.7-14 The inductor L and the capacitor C in Fig. P1.7-14 are nonlinear, which makes the circuit nonlinear. The remaining three elements are linear. Show that the output $y(t)$ of this nonlinear circuit satisfies the linearity conditions with respect to the input $x(t)$ and the initial conditions (all the initial inductor currents and capacitor voltages).

1.7-15 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, determine which are causal and which are noncausal.

(a) $y(t) = x(t-2)$
(b) $y(t) = x(-t)$
(c) $y(t) = x(at) \qquad a > 1$
(d) $y(t) = x(at) \qquad a < 1$

1.7-16 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, determine which are invertible and which are noninvertible. For the invertible systems, find the input–output relationship of the inverse system.

(a) $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$
(b) $y(t) = x^n(t)$, $x(t)$ real, n integer
(c) $y(t) = \dfrac{dx(t)}{dt}$
(d) $y(t) = x(3t-6)$
(e) $y(t) = \cos[x(t)]$
(f) $y(t) = e^{x(t)}, \quad x(t)$ real

1.7-17 Figure P1.7-17 displays an input $x_1(t)$ to a linear time-invariant (LTI) system H, the corresponding output $y_1(t)$, and a second input $x_2(t)$.

(a) Bill suggests that $x_2(t) = 2x_1(3t) - x_1(t-1)$. Is Bill correct? If yes, prove it. If not, correct his error.
(b) Bill wants to know the output $y_2(t)$ in response to the input $x_2(t)$. Provide him with an expression for $y_2(t)$ in terms of $y_1(t)$. Use MATLAB to plot $y_2(t)$.

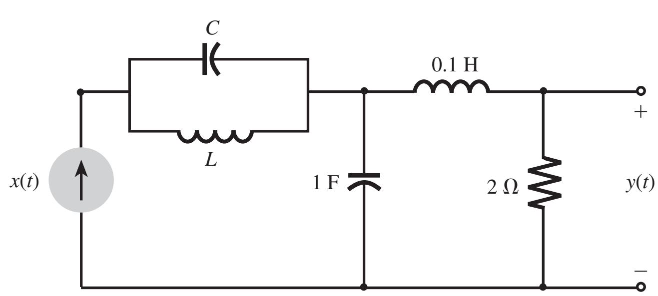

Figure P1.7-14

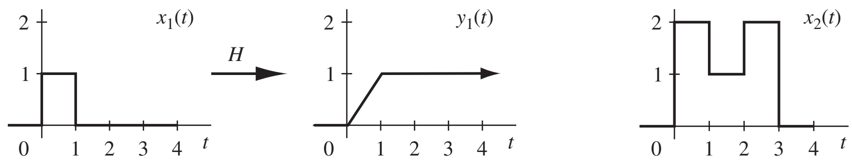

Figure P1.7-17

1.7-18 A linear time-invariant system H acts on input $x(t) = u(t-0.5) - u(t-1.5)$ to produce output $y(t) = H\{x(t)\} = 0.5u(t) + 0.5u(t-1) - u(t-2)$.

(a) Is it possible that the system is causal? Explain your answer. If not causal, determine the shift necessary to make the system causal.

(b) Is it possible that the system is memoryless? Explain your answer.

(c) Suppose the output $y(t)$ is applied to an identical system H to produce output

$$z(t) = H\{y(t)\} = H\{H\{x(t)\}\}$$

If possible, determine and sketch $z(t)$. If not possible, explain why $z(t)$ cannot be determined using the information given.

1.8-1 For the circuit depicted in Fig. P1.8-1, find the differential equations relating outputs $y_1(t)$ and $y_2(t)$ to the input $x(t)$.

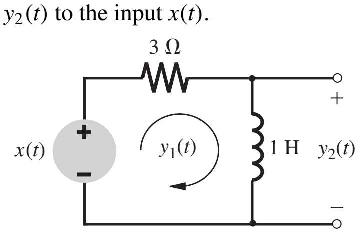

Figure P1.8-1

1.8-2 For the circuit depicted in Fig. P1.8-2, find the differential equations relating outputs $y_1(t)$ and $y_2(t)$ to the input $x(t)$.

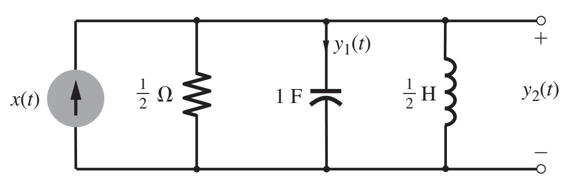

Figure P1.8-2

1.8-3 A simplified (one-dimensional) model of an automobile suspension system is shown in Fig. P1.8-3. In this case, the input is not a force but a displacement $x(t)$ (the road contour). Find the differential equation relating the output $y(t)$ (auto body displacement) to the input $x(t)$ (the road contour).

1.8-4 A field-controlled dc motor is shown in Fig. P1.8-4. Its armature current i_a is maintained constant. The torque generated by this motor is proportional to the field current i_f (torque= $K_f i_f$). Find the differential equation relating the output position θ to the input voltage $x(t)$. The motor and load together have a moment of inertia J.

1.8-5 Water flows into a tank at a rate of q_i units/s and flows out through the outflow valve at a rate of q_0 units/s (Fig. P1.8-5). Determine the equation relating the outflow q_0 to the input q_i. The outflow rate is proportional to the head h. Thus $q_0 = Rh$, where R is the valve resistance. Determine also the differential equation relating the head h to the input q_i. [Hint: The net inflow of water in time Δt is $(q_i - q_0)\Delta t$. This inflow is also $A\Delta h$, where A is the cross section of the tank.]

1.8-6 Consider the circuit shown in Fig. P1.8-6, with input voltage $x(t)$ and output currents $y_1(t)$, $y_2(t)$, and $y_3(t)$.

(a) What is the order of this system? Explain your answer.

(b) Determine the matrix representation for this system.

(c) Use Cramer's rule to determine the output current $y_3(t)$ for the input voltage $x(t) = [2 - |\cos(t)|]u(t-1)$.

1.10-1 Write state equations for the parallel *RLC* circuit in Fig. P1.8-2. Use the capacitor voltage q_1 and the inductor current q_2 as your state variables.

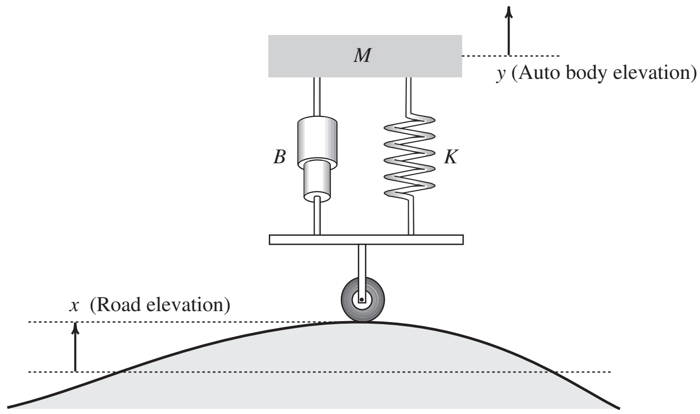

Figure P1.8-3

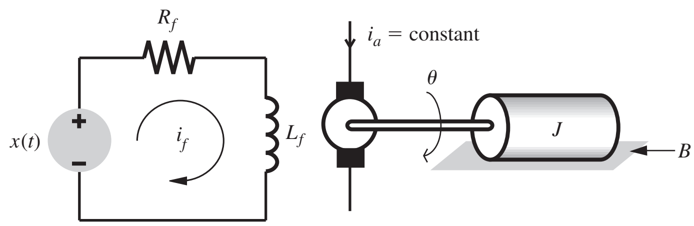

Figure P1.8-4

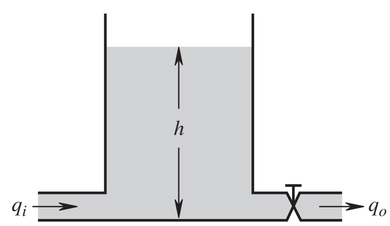

Figure P1.8-5

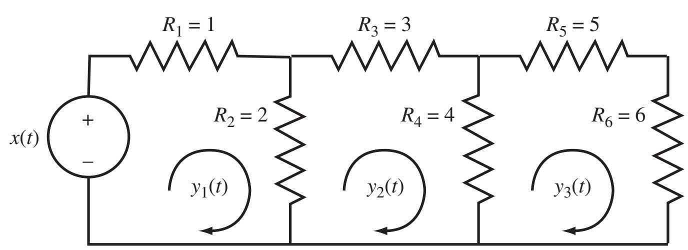

Figure P1.8-6

Show that every possible current or voltage in the circuit can be expressed in terms of q_1, q_2 and the input $x(t)$.

1.10-2 Write state equations for the third-order circuit shown in Fig. P1.10-2, using the inductor currents q_1, q_2 and the capacitor voltage q_3 as state variables. Show that every possible voltage or current in this circuit can be expressed as a linear combination of q_1, q_2, q_3, and the input $x(t)$. Also, at some instant t, it was found that

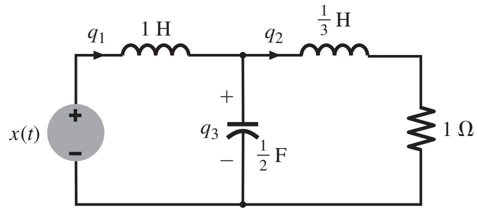

Figure P1.10-2

$q_1 = 5$, $q_2 = 1$, $q_3 = 2$, and $x = 10$. Determine the voltage across and the current through every element in this circuit.

1.11-1 Provide MATLAB code and output that plots the odd portion $x_o(t)$ of the function $x(t) = 2^{-t}\cos(2\pi t)u(t-\pi)$ over a suitable-length interval using a suitable number of points.

1.11-2 Provide MATLAB code and output that plots the even portion $x_e(t)$ of the function $x(t) = 2^{-t/2}\cos(4\pi t)u(t-0.5)$ over a suitable t using $\Delta t = 0.002$ second between points.

1.11-3 Define $x(t) = e^{t(1+j2\pi)}u(-t)$ and $y(t) = \mathrm{Re}\left\{2x\left(\frac{-5-t}{2}\right)\right\}$.

(a) Use MATLAB to plot $\mathrm{Re}\{x(t)\}$ versus $\mathrm{Im}\{x(at)\}$ for $a = 0.5$, 1, and 2 and $-10 \leq t \leq 10$. How important is the scale factor a on the shape of the resulting figure?

(b) Use MATLAB to plot $y(t)$ over $-10 \leq t \leq 10$. Analytically determine the time t_0 where $y(t)$ has a jump discontinuity. Verify your calculation of t_0 using the plot of $y(t)$.

(c) Use MATLAB and numerical integration to compute the energy E_x of signal $x(t)$.

(d) Use MATLAB and numerical integration to compute the energy E_y of signal $y(t)$.

1.11-4 Consider the signal $x(t) = u(\frac{t}{2}+1) - u(t-1) - \delta(\frac{t}{2})$. Define $y(t) = \int_{-\infty}^{t-3} x(\tau)\,d\tau$ and $z(t) = \int_t^{\infty} x(\tau)\,d\tau$.

(a) Using MATLAB, accurately plot $y(t)$.

(b) Using MATLAB, accurately plot $z(t)$.

(c) Using MATLAB, accurately plot $w(t) = \frac{d}{dt}\left[y(t)+z(t)\right]$.